Computation for Analytics

DIANE WOODBRIDGE, PH.D

Announcement

Reminder

- No late/re-submission allowed.
- If you want to ask questions about the class, please post on Piazza instead of DM.

Get to know each other - Please make sure to sign up by August 10th

Assignment 2 (Group HW) - Due by Thursday, July 20th 9 PM

Peer Review for HW1_2 - Due by Monday, July 24th Midnight

Quiz 1 - Today 1pm - 2pm : Make sure to be here by 12:50 for attendance check

Quiz 2 - Friday, July 14th 1pm - 2pm

- Study list and practice quiz are available on Canvas.

Please provide feedback - What did you like/dislike so far?

Nobody has responded yet.

Hang tight! Responses are coming in.

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Day 3 - Contents

Python Data Types

Conditional Statements

Loop

Iterator vs Generator

Iterator

- An object that can be iterated upon.
- Sequence data types are iterable container that has an iterator.

```
iter(your_sequence)
next(your_sequence)
```

Generator

- Can be used to **create an iterator object** with the iterator itself and the next item.
- Need to use **`yield`** as a key word
- It's lazily evaluated and improves memory usage.

ex-generator.ipynb

https://wiki.python.org/moin/Generators

Questions for Day 3 Topics?

Nobody has responded yet.

Hang tight! Responses are coming in.

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What is the output of the following code?

```
a = (1,2,3)
a[2] = 1
print(a)
```

- (1, 2, 3)
- (1, 2, 1)
- (1, 1, 3)
- 1
- Error

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What is the output of the following code?

```
a = (1,2,3)
a[2] = 1
print(a)
```

(1, 2, 3) 0%

(1, 2, 1) 0%

(1, 1, 3) 0%

1 0%

Error 0%

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```
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```

(1, 2, 3) 0%

(1, 2, 1) 0%

(1, 1, 3) 0%

1 0%

Error 0%

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What is the output of the following code?

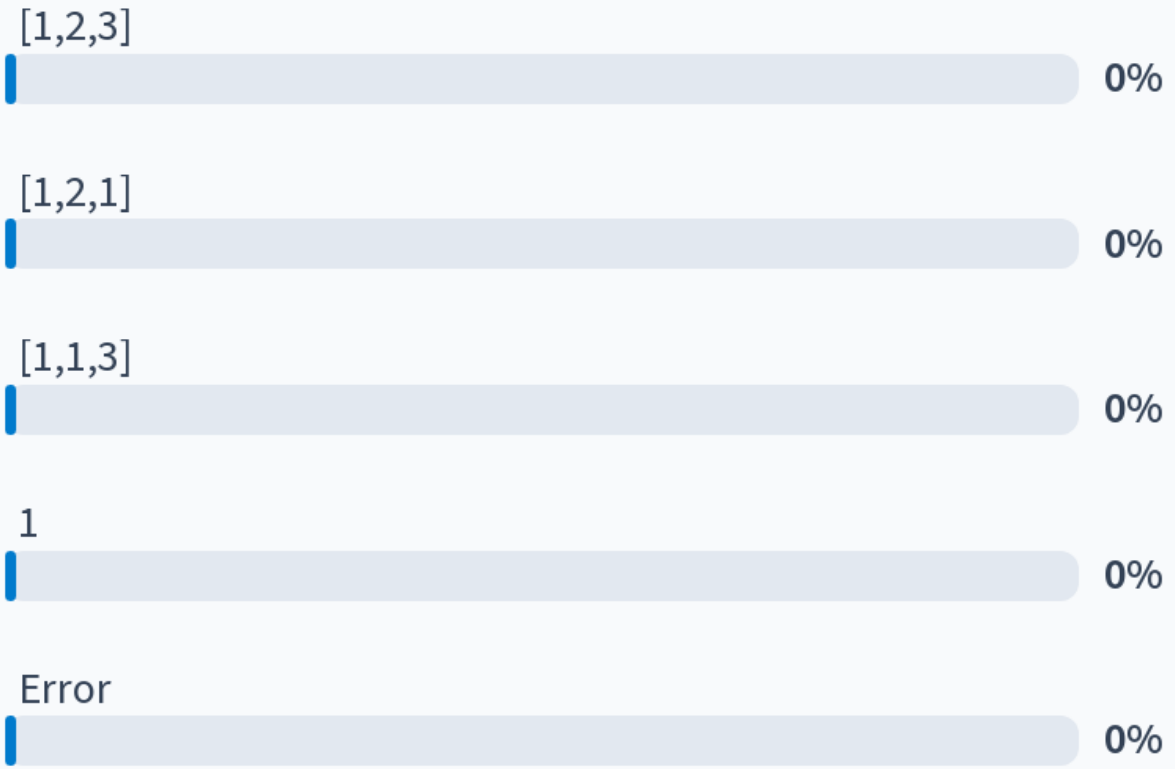
```
a = [1,2,3]
a[2] = 1
print(a)
```

- [1,2,3]
- [1,2,1]
- [1,1,3]
- 1
- Error

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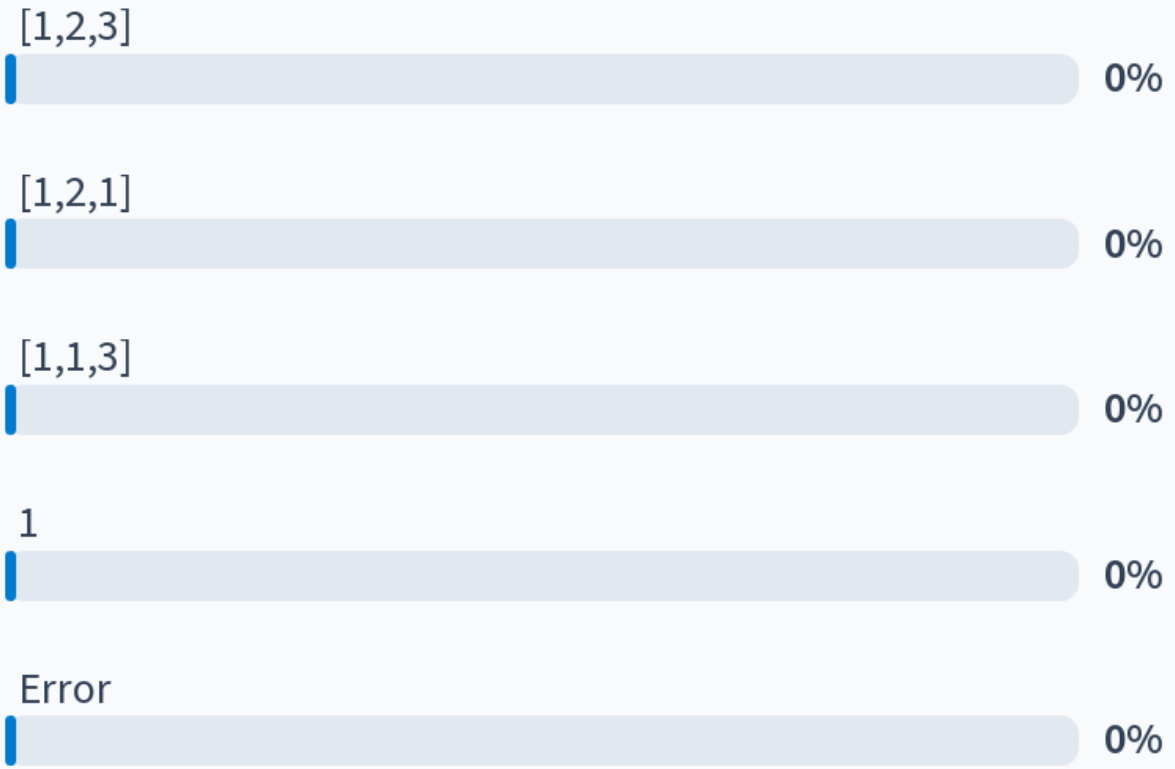
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a = [1,2,3]
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```



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print(a)
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What is the output of the following code?

```
a = [1,2,3]
b = a
a = [1,3,4]
print(b)
```

- [1, 2, 3]
- [1, 3, 4]
- a
- Error
- None of the above

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What is the output of the following code?

```
a = [1,2,3]
b = a
a = [1,3,4]
print(b)
```

[1, 2, 3] 0%

[1, 3, 4] 0%

a 0%

Error 0%

None of the above 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What is the output of the following code?

```
a = [1,2,3]
b = a
a = [1,3,4]
print(b)
```

[1, 2, 3] 0%

[1, 3, 4] 0%

a 0%

Error 0%

None of the above 0%

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What is the output of the following code?

```
j = 0
output = []
for i in range(5):
    output.append(j)
    j = i + 1
else:
    output.append(j)
print(output)
```

- [0, 1, 2, 3, 4]
- [1, 2, 3, 4, 5]
- [0, 1, 2, 3, 4, 5]
- [1, 2, 3, 4, 5, 6]
- None of the above

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- [0, 1, 2, 3, 4] 0%
- [1, 2, 3, 4, 5] 0%
- [0, 1, 2, 3, 4, 5] 0%
- [1, 2, 3, 4, 5, 6] 0%
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- [0, 1, 2, 3, 4] 0%
- [1, 2, 3, 4, 5] 0%
- [0, 1, 2, 3, 4, 5] 0%
- [1, 2, 3, 4, 5, 6] 0%
- None of the above 0%

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Choose how many lines would be printed after executing the following code?

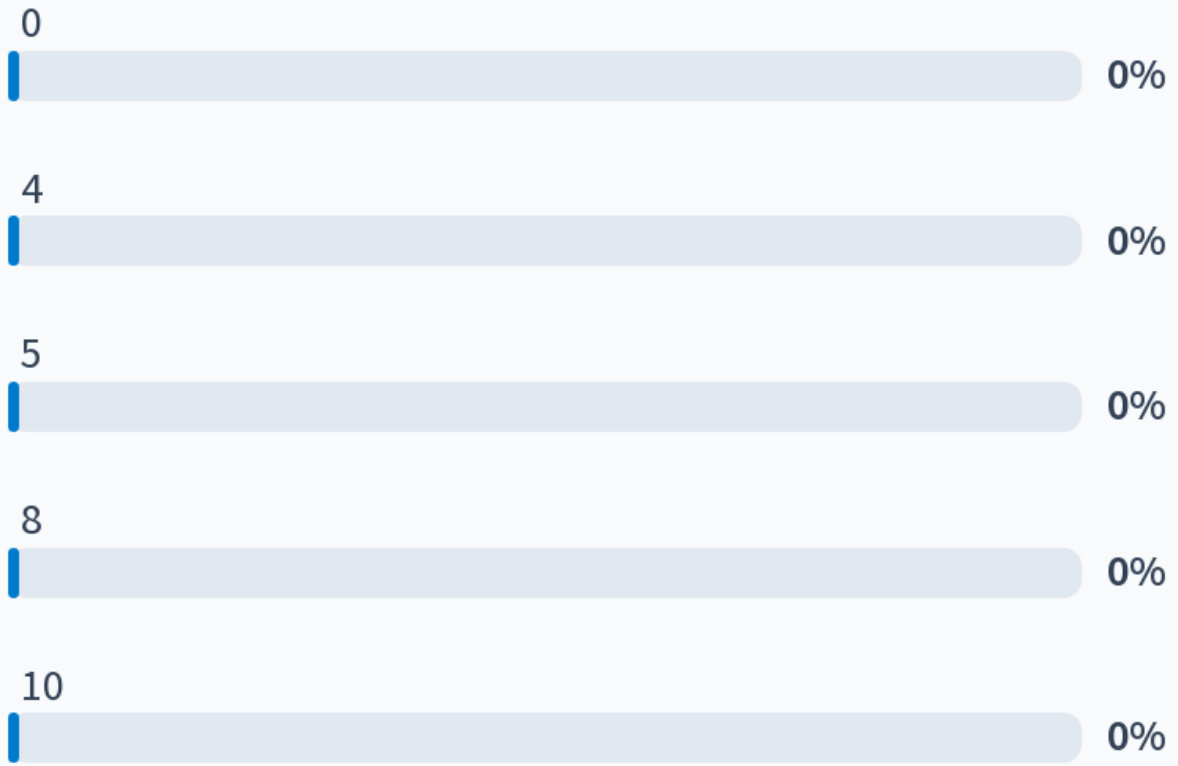
```
i = 0
while i < 10:
    i += 1
    if i < 5:
        pass
        print("after pass")
        continue
        print("after continue")
```

- 0
- 4
- 5
- 8
- 10

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Choose how many lines would be printed after executing the following code?

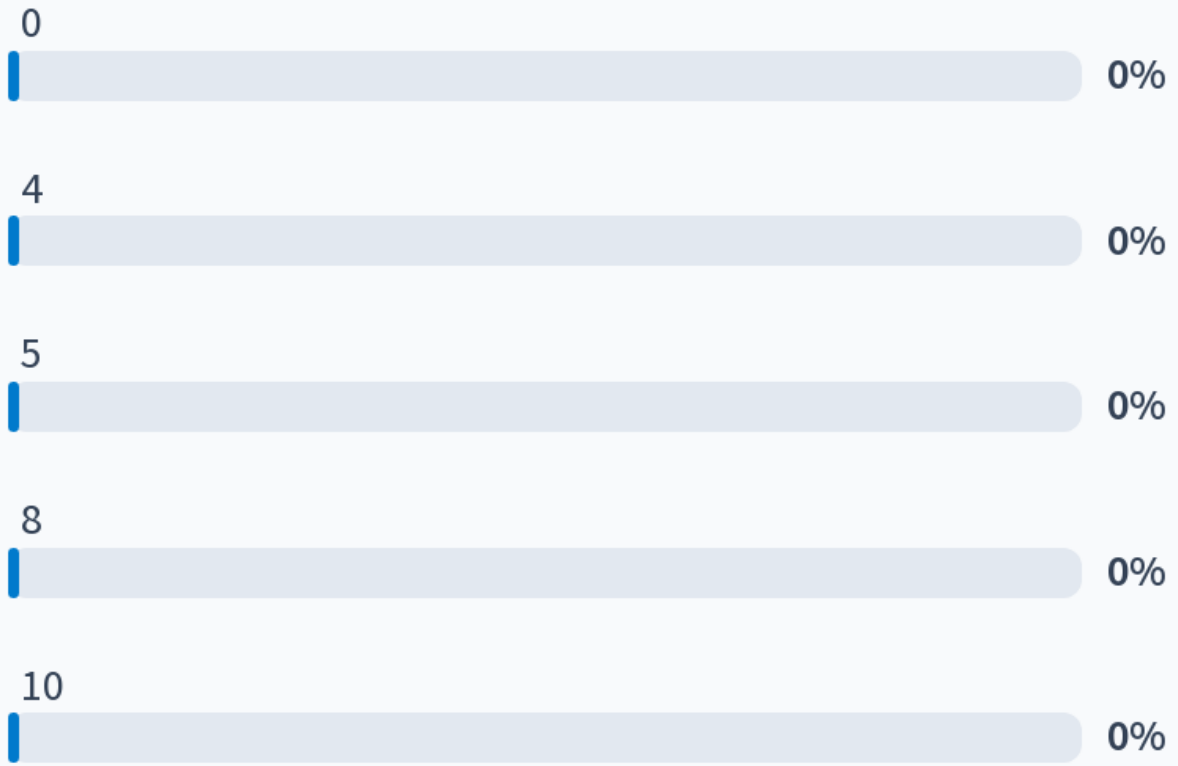
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```



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Day 4 - Contents

Python File I/O

Function

Data Aliasing

pytest

main()

In general, most programming languages require main() function which is an entry point.

- Although Python does not require main(), define it helps developers understand to find the starting point and improves readability.

```python
def greet():
    print("Welcome everyone")

greet()

def introduce(name: str):
    print(f'My name is {name}')

introduce("Diane")
```

✓ 0.0s

```
Welcome everyone
My name is Diane
```

VS

```python
def greet():
    print("Welcome everyone")

def introduce(name: str):
    print(f'My name is {name}')

def main():
    greet()
    introduce("Diane")

if __name__=='__main__':
    main()
```

✓ 0.0s

```
Welcome everyone
My name is Diane
```

Decorator

Syntax

```python
import time

def decorator_name(func):  # the decorator takse the function
    def wrapper_func(*args, **kargs): # the inner function takes the argument of func and add extra on func
        # args and kargs are arbitary or arbitary keyword arguments given to func
        before = time.time()  # add some additional behaviors
        func(*args, **kargs)
        print(f'Took {time.time() - before} second') # add some additional behaviors
    return wrapper_func

@decorator_name
def hello_world(**kargs):
    print(f"hello word - {kargs['name']}")
```

ex-decorator.ipynb

Decorator

Benefits

- You can easily update the original function's behaviors without changing the function itself by wrapping a function.
- By simply adding @ in front of the decorator's name, you can simply use it.
- Also, you can reuse a decorator for different functions.

Questions for Day 4 Topics?

Nobody has responded yet.

Hang tight! Responses are coming in.

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What is the command to read all the following lines after reading the first line by f.readline()

- f.readline()
- f.readlines()
- f.load()
- f.loads()
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What is the command to read all the following lines after reading the first line by f.readline()

f.readline() 0%

f.readlines() 0%

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f.loads() 0%

None of the above 0%

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None of the above 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

For a function call, generate_name("Diane", "Woodbridge", "Dr"), which one of the following does not throw an error?

```python
# a
def generate_name(first, last):
    print(f'{first} {last}')
```

```python
# b
def generate_name(*args):
    print(f'{args[0]} {args[1]}')
```

```python
# c
def generate_name(*kargs):
    print(f'{kargs["first"]} {kargs["last"]}')
```

```python
# d
def generate_name(**kargs):
    print(f'{kargs["first"]} {kargs["last"]}')
```

a **(A)**

B **(B)**

C **(C)**

D **(D)**

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def generate_name(*kargs):
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```

```python
# d
def generate_name(**kargs):
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```

(A) a 0%

(B) B 0%

(C) C 0%

(D) D 0%

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```

(A) a 0%

(B) B 0%

(C) C 0%

(D) D 0%

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In order to have the last output, which one has the correct definition for "decorator_name()"?

```python
def decorator_name(func):
    def wrapper(*args):
        print("Welcome", end=', ')
        func(*args)
    return wrapper

# b
def decorator_name(func):
    def wrapper(*args):
        print("Welcome", end=', ')
        func(*args)

# c
def decorator_name(func(*args)):
    def wrapper():
        print("Welcome", end=', ')
        func(*args)
    return wrapper
```

✓ 0.0s

```python
@decorator_name
def generate_name(*args):
    print(f'{args[0]} {args[1]}')
```

✓ 0.0s

```python
generate_name("Diane", "Woodbridge", "Dr")
```

✓ 0.0s

A

B

C

None of the above

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        func(*args)
    return wrapper
```

✓ 0.0s

```python
@decorator_name
def generate_name(*args):
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```

✓ 0.0s

```python
generate_name("Diane", "Woodbridge", "Dr")
```

✓ 0.0s

```
Welcome, Diane Woodbridge
```

A 0%

B 0%

C 0%

None of the above 0%

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✓ 0.0s

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```

✓ 0.0s

```python
generate_name("Diane", "Woodbridge", "Dr")
```

✓ 0.0s

```
Welcome, Diane Woodbridge
```

A 0%

B 0%

C 0%

None of the above 0%

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